

Commands & API Endpoints

Overview

The Receipt Validator backend is a FastAPI application that accepts receipt images and PDFs, extracts structured data using an AI vision model, validates the math and authenticity of the receipt, and returns a verdict for each file. This document covers all commands to run the project and all available API endpoints with their purpose, usage, and guardrails.

Commands

Backend

Install dependencies

```
cd backend
pip install -r requirements.txt
```

Installs all Python dependencies including FastAPI, Uvicorn, Pillow, PyMuPDF, pdf2image, and the OpenAI-compatible SDK.

Run locally

```
uvicorn main:app --reload --port 8001
```

Starts the backend server on <http://localhost:8001>. The --reload flag enables hot-reloading on file changes (development only).

Run with Docker locally

```
docker build -t receipt-validator .
docker run -p 8001:8001 --env OPENROUTER_API_KEY=sk-or-... receipt-validator
```

Builds and runs the Docker image locally. Useful for testing the production environment before deploying.

Frontend

Install dependencies

```
cd frontend
npm install
```

Run locally

```
npm run dev
```

Starts the Vite dev server at <http://localhost:5173>.

Build for production

```
npm run build
```

Outputs static files to frontend/dist/.

curl Commands

Validate a single receipt image

```
curl -X POST https://ai-receipt-validator.onrender.com/api/v1/validate \
-F "files=@receipt.jpg;type=image/jpeg" | python3 -m json.tool
```

Validate a single PDF

```
curl -X POST https://ai-receipt-validator.onrender.com/api/v1/validate \
-F "files=@receipt.pdf;type=application/pdf" | python3 -m json.tool
```

Validate multiple files (up to 3)

```
curl -X POST https://ai-receipt-validator.onrender.com/api/v1/validate \
-F "files=@receipt1.jpg;type=image/jpeg" \
-F "files=@receipt2.png;type=image/png" \
-F "files=@receipt3.pdf;type=application/pdf" | python3 -m json.tool
```

Health check

```
curl https://ai-receipt-validator.onrender.com/health
```

API Endpoints

POST /api/v1/validate

Purpose

Accepts one or more receipt files, extracts structured data using an AI vision model, validates the math, detects fraud signals, and returns a verdict for each file.

Base URL: <https://ai-receipt-validator.onrender.com>

Request

- Method: POST
- Content-Type: multipart/form-data
- Field name: files (repeat for multiple files)
- Max files: 3 per request

Accepted File Types

MIME Type	Format
image/jpeg	JPG / JPEG
image/png	PNG
image/webp	WebP
image/tiff	TIFF

image/bmp	BMP
application/pdf	PDF (native text or scanned)

Response Fields

Field	Type	Description
is_receipt	bool	Whether the uploaded file is a receipt
merchant	string / null	Merchant name extracted from the receipt
date	string / null	Receipt date in ISO format (YYYY-MM-DD)
currency	string / null	Currency code (e.g. USD, PHP, EUR)
line_items	array	List of { description, total_price } objects
subtotal	number / null	Subtotal as printed on the receipt
tax_total	number / null	Tax amount as printed on the receipt
tip	number / null	Tip amount if present
total	number / null	Final total as printed on the receipt
is_handwritten	bool	Whether the receipt appears handwritten
confidence	string	AI confidence: high, medium, or low
fraud_signals	array	List of { description, severity } objects
math_valid	bool / null	Whether subtotal + tax + tip = total (+/-0.02)
discrepancy	number / null	Absolute difference between computed and printed total
calculated_sub_total	number / null	Sum of all line item prices
line_items_sum_valid	bool / null	Whether calculated_sub_total is approximately equal to subtotal
fraud_signal_rating	string / null	Worst fraud signal severity: high, mid, or low
is_accepted	bool / null	Final accept/reject verdict
warning	string / null	Human-readable warning, pipe-separated if multiple

Fraud Signal Severity

Severity	Meaning
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high	Totals appear fabricated or altered — strong indicator of fraud
mid	Math does not add up — possible error or manipulation
low	Missing or illegible data — reduces confidence only

Guardrails

Rule	Behaviour
Max 3 files per request	Returns 400 if more than 3 files are uploaded
Unsupported file type	Returns 400 with a descriptive error message
Unreadable image	Returns 400 — file may be corrupt or unsupported
Empty AI response	Returns 500 with "Model returned an empty response"
is_receipt = false	Returns is_accepted = false with a high fraud signal
total = null	Skips math validation, adds warning
confidence = low	Adds "Please confirm manually" to warning
line_items = null from AI	Normalised to empty list before validation
fraud_signals = null from AI	Normalised to empty list before validation

Error Responses

```
{ "detail": "Maximum 3 files allowed per request." }
{ "detail": "Unsupported file type: image/gif. Allowed: jpg, png, webp, tiff, bmp, pdf" }
{ "detail": "Could not read image file 'receipt.webp'. The file may be corrupt." }
```

GET /health

Purpose

Health check endpoint. Used by Render and other hosting platforms to verify the service is running.

Request

```
curl https://ai-receipt-validator.onrender.com/health
```

Response

```
{ "status": "ok" }
```

Guardrails

- No authentication required
- Always returns 200 if the server is running
- Set this as the health check path in Render dashboard (/health) to avoid false 404 failures on /

Notes

- The backend is stateless — no data is stored between requests
- All math validation uses Python Decimal arithmetic for precision
- Images are resized to max 1024px before being sent to the AI model to reduce latency
- PDFs with embedded text are sent as text; scanned PDFs are converted to images at 150 DPI
- The AI model used is nvidia/nemotron-nano-12b-v2-vl:free via OpenRouter